Introduction to Kanban

Digital T-Level – Software Testing

Today

- Definition of Lean
- What is Kanban

Lean – Eliminating Waste

- Lean project management was born in the manufacturing industry.

- The term “Lean Manufacturing” was adopted in 1988, but…
- …is has roots dating back to the 1930’s from Toyota.

- Lean is defined as a mindset that comes with values and principles.
- The mindset is often referred to as “Lean Thinking”

Lean Values

- Eliminate waste
- Amplify learning
- Decide as late as possible
- Deliver as fast as possible
- Empower the team
- Build integrity in
- See the whole

Kanban

- Kanban was developed by Toyota.
- Using the lean process of “Just-in-time”
- Toyota used Kanban – which is Japanese for “signboard” – to control manufacturing processes.
- Kanban uses a boar (physical or digital) to track workflow of a project.
- There are variations on the layout of the board, but generally they track some form of
  - To do
  - Doing
  - Done

Kanban Board

- Examples of Kanban boards:

- http://bit.ly/KanbanBoardExamples

Kanban and Software Development

- Kanban itself is not a software development methodology.
- Kanban is about helping teams improve the way that they build software

- Kanban teams have a clear picture of:
  - The actions required to build a software product.
  - How they interact with the rest of the company
  - Where they run into waste caused by inefficiency
  - How to improve over time by removing the root cause of any inefficiency.

Kanban Foundational Principles

- Kanban has both foundational principles and core practices.

- Foundational principles:
  - Start with what you do know
  - Agree to pursue incremental, evolutionary change
  - Initially, respect current roles, responsibilities & job titles

*https://digital.ai/resources/agile-101/what-is-kanban

Kanban Core Principles

- Adopt after following the foundational principles:
  - Visualise
  - Limit work in progress (WIP)
  - Manage flow
  - Male process policies explicit
  - Implement feedback loops
  - Improve collaboratively, explore experimentally

Kanban and Agile

- Kanban is **lean not agile**, but they share common ideas.
- “Teams that use Kanban start by understanding the way that they currently build software, and make improvements over time.”*

- Kanban is often adopted alongside Scrum…
- …to the point that there is Scrumban!

*Learning Agile, Andrew Stellman & Jennifer Greene, 2015, O’Reilly Publishing

Kanban and Software Testing

- Testing takes place after the development of a feature is complete.
- Testing is post development and pre-production.
- Work is completed ‘just-in-time’, so testing also follows this principle.

- Kanban works well with agile processes (like Scrum) as it welcomes changes in scope…
- …and includes continuous delivery.

Kanban Advantages and Disadvantages

- In the activity, you will need to compile a list of both positive and negative points of adopting Kanban when developing software projects.

- When completing this activity, remember to relate it to software testing (not manufacturing).